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CNN-Based Brain Tumor Classification Using MRI Imaging

CAP5626 ARTIFICIAL INTELLIGENCE

SPRING 2026

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Abstract

Early detection of brain tumors plays a critical role in improving patient outcomes. However, traditional diagnosis relies heavily on manual interpretation of magnetic resonance imaging scans, which can be time-consuming and subject to human error. With the increasing availability of medical imaging data, there is a growing need for automated systems that can assist healthcare professionals. This project presents a convolutional neural network designed to classify brain MRI images into four categories: glioma, meningioma, pituitary tumor, and no tumor. The model was trained using a labeled dataset of MRI images and evaluated using accuracy, loss, and classification metrics. The results show that the model achieved a test accuracy of approximately 82 percent, with strong performance in identifying non-tumor and pituitary cases. While the model demonstrates promising results, certain limitations highlight opportunities for further improvement. This study emphasizes the potential of deep learning as a supportive tool in medical diagnostics.

1. Introduction & Problem Motivation

Brain tumors are serious medical conditions that require timely and accurate diagnosis. Magnetic resonance imaging, commonly known as MRI, is one of the most widely used techniques for detecting abnormalities in the brain. MRI scans provide detailed images of brain structures, which makes them useful for identifying tumors and other medical concerns. However, interpreting these images requires skill, experience, and careful review by trained medical professionals.

The motivation for this project comes from the need to improve efficiency and consistency in medical image analysis. Radiologists often work with large numbers of scans, and manual review can be time-consuming. In some cases, diagnosis may also vary depending on the experience of the person reviewing the image. While artificial intelligence is not a replacement for doctors, it can support the diagnostic process by helping identify patterns in medical images.

Deep learning has become especially useful in image classification tasks because it can learn important features directly from image data. Convolutional neural networks are designed to recognize visual patterns such as edges, shapes, and textures. This makes them a strong choice for brain MRI classification. In this project, a convolutional neural network is used to classify MRI images into four categories: glioma, meningioma, pituitary tumor, and no tumor.

1.1 Problem Statement

The problem addressed in this project is the development of an automated system capable of classifying brain MRI images into distinct tumor categories. Manual analysis of MRI scans can be time-consuming

and may lead to inconsistencies due to human interpretation. The objective of this project is to evaluate whether a convolutional neural network can effectively learn features from MRI images and accurately classify them into glioma, meningioma, pituitary tumor, and no tumor categories. The goal is to improve diagnostic support by providing a reliable and efficient classification system.

1.2 Figures and Tables

This report includes visual evidence to support the model evaluation. The accuracy and loss graphs show how the model performed during training and validation. The confusion matrix shows how the model classified each tumor category on the test dataset. These figures are important because they provide more detail than accuracy alone. They help show where the model performed well and where misclassifications occurred.

2. Literature Review

Recent research has shown that deep learning methods are effective for medical image classification. Convolutional neural networks are commonly used in medical imaging because they can automatically extract important visual features from raw images. Unlike traditional machine learning methods, CNN models do not require manual feature extraction. This makes them useful for complex image-based problems such as brain tumor classification.

Studies published in recent years have shown that CNN-based models can achieve strong results when classifying brain MRI images. These models are able to identify patterns in tumor shape, texture, and location that may be difficult to detect manually. Research also shows that preprocessing steps such as resizing, normalization, and organized class labeling improve model performance and training stability. Recent studies such as Ismael et al. (2020) and Rehman et al. (2020) demonstrate that convolutional neural networks achieve strong performance in brain tumor classification tasks using MRI images.

Although many advanced studies use complex architectures or pretrained models, this project focuses on a simpler CNN architecture. The purpose is to show that a practical model can still produce useful results while remaining easier to understand, train, and reproduce. This approach is appropriate for a graduate capstone because it connects current research with a working technical implementation. These findings support the use of convolutional neural networks as a reliable approach for automated brain tumor classification in medical imaging.

3. Background

Artificial intelligence is a broad field focused on building systems that can perform tasks requiring human-like reasoning, learning, or decision-making. In this project, the AI method used is deep learning, which is a subset of machine learning that uses neural networks with multiple layers. Deep learning is especially powerful for image analysis because it can learn patterns directly from large amounts of image data.

A convolutional neural network is a type of deep learning model designed for image recognition. CNNs use convolutional layers to scan images and detect visual features. Early layers often learn simple patterns such as edges and corners, while deeper layers learn more complex patterns. Pooling layers reduce image dimensions and help the model focus on the most important features. Fully connected layers then use these learned features to make a final classification.

For this project, the CNN model is trained to classify MRI images into four categories. The four categories are glioma, meningioma, pituitary tumor, and no tumor. Since this is a multi-class classification problem, the model uses a SoftMax output layer and categorical cross-entropy loss.

4. Methodology

The methodology for this project involved preparing the MRI dataset, building a convolutional neural network, training the model, and evaluating its performance. The project was completed in Google Colab using TensorFlow and Keras.

The model architecture included three convolutional layers. The first layer used 32 filters, the second used 64 filters, and the third used 128 filters. Each convolutional layer was followed by a max pooling layer. This design allowed the model to learn increasingly complex image features while reducing the size of the feature maps.

After the convolutional and pooling layers, the model used a flattening layer to convert the extracted features into a format that could be passed into dense layers. A dense layer with 128 neurons was used to learn higher-level patterns from the extracted features. A dropout layer was included to reduce overfitting. The final output layer used four neurons with a SoftMax activation function, representing the four image classes.

The model was compiled using the Adam optimizer and categorical cross-entropy loss. Accuracy was used as the main evaluation metric. The model was trained for five epochs using a batch size of 32.

5. Dataset and Implementation

The dataset used in this project contained brain MRI images divided into four classes: glioma, meningioma, no tumor, and pituitary tumor. The data was organized into Training and Testing folders. The training folder contained 5,600 total images before validation splitting. Using an 80 percent training and 20 percent validation split, 4,480 images were used for training, and 1,120 images were used for validation. The testing folder contained 1,600 images.

All images were resized to 224 by 224 pixels so they could be used as consistent input for the CNN model. The image pixel values were rescaled by dividing by 255, which normalized the data to a range between 0 and 1. This preprocessing step helped improve training stability.

The implementation was completed in Google Colab. TensorFlow and Keras were used to build, train, and evaluate the model. The ImageDataGenerator class was used to load the images from the directory structure, apply normalization, and create batches for training, validation, and testing.

6. Model and Analysis

The CNN model showed steady improvement during training. In the first epoch, the model achieved a training accuracy of approximately 68.46 percent and a validation accuracy of approximately 76.52 percent. By the fifth epoch, training accuracy increased to approximately 93.75 percent, and validation accuracy reached approximately 91.07 percent. As shown in Figure 1, the model demonstrates a steady increase in both training and validation accuracy across epochs, indicating effective learning.

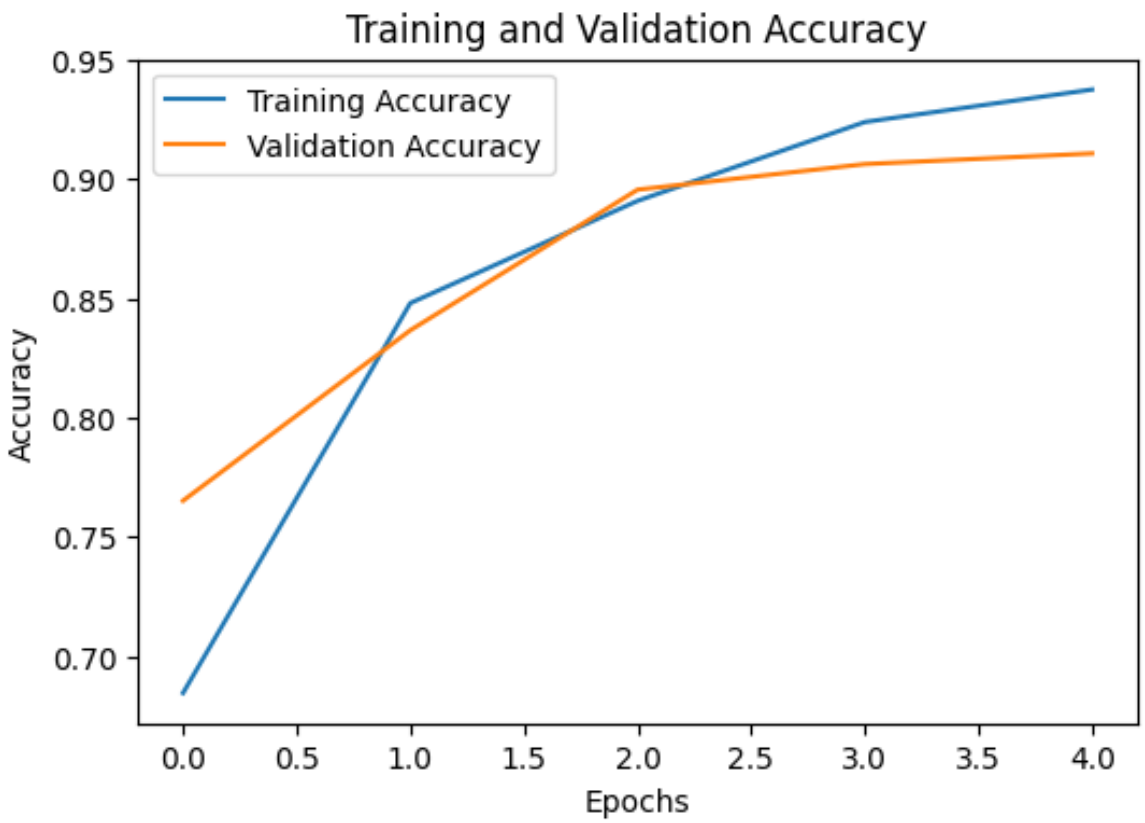

Figure 1. Training and validation accuracy across epochs, showing the progressive improvement of the convolutional neural network during the learning process. The graph illustrates how both training and validation accuracy increase over time, indicating effective feature learning and strong generalization performance with minimal signs of overfitting.

The loss values also improved throughout training. Training loss decreased from approximately 0.7598 in the first epoch to approximately 0.1749 in the fifth epoch. Validation loss decreased from approximately 0.5054 in the first epoch to approximately 0.2528 in the final epoch. This pattern shows that the model was learning effectively. Figure 2 illustrates the decrease in training and validation loss, showing that the model is minimizing classification error over time.

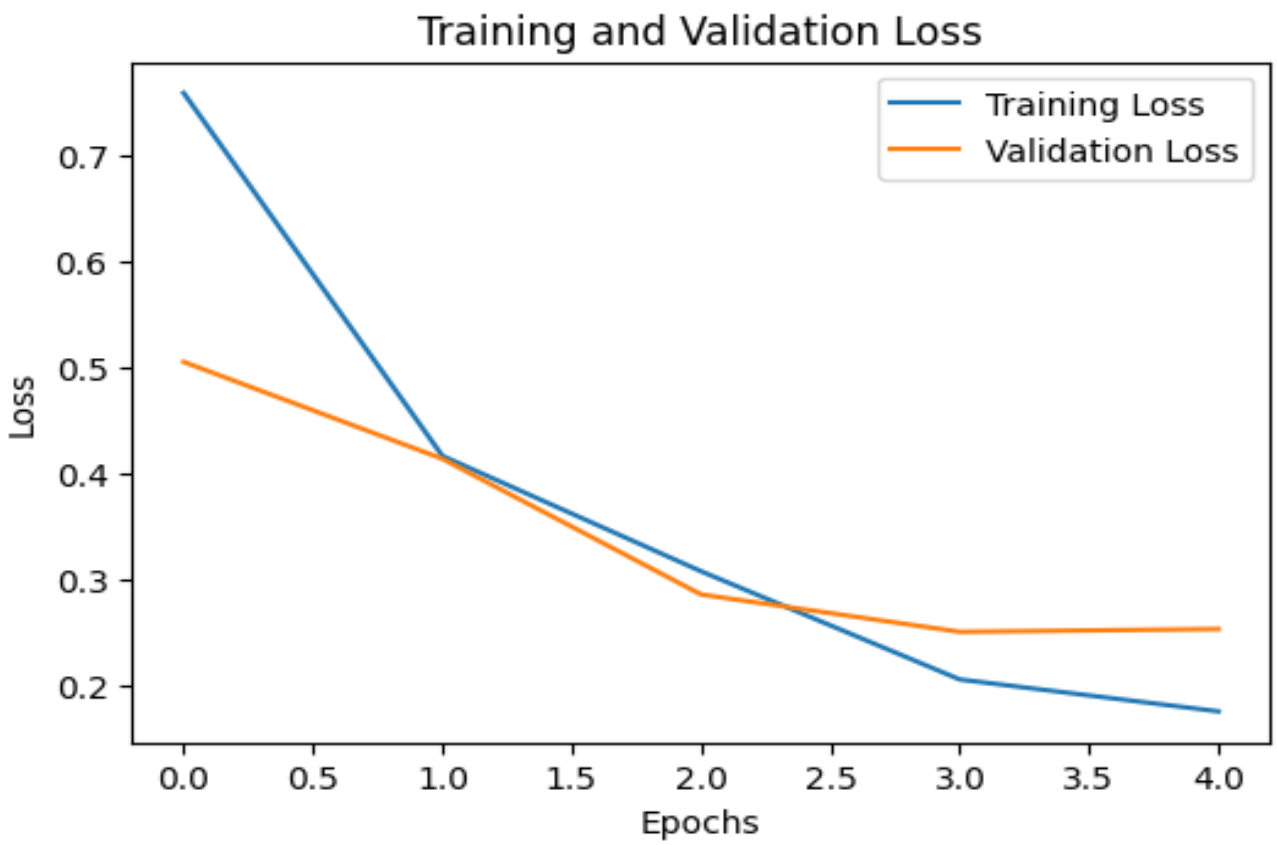

Figure 2. Training and validation loss across epochs, demonstrating the model's ability to minimize classification error throughout the training process. The consistent decrease in both training and validation loss suggests stable learning behavior and supports the model's ability to converge effectively.

The validation accuracy remained close to the training accuracy, which suggests that the model did not severely overfit during the five training epochs. The use of dropout and validation monitoring helped support generalization.

7. Results

The convolutional neural network demonstrated strong performance when evaluated on unseen test data. The model achieved an overall test accuracy of approximately 82 percent, which indicates that it was able to correctly classify the majority of brain MRI images across all categories. This level of performance suggests that the model successfully learned meaningful features from the training data and was able to apply those patterns to new images.

During training, the model showed steady improvement across all five epochs. Training accuracy increased from approximately 68 percent in the first epoch to over 93 percent by the final epoch. Similarly, validation accuracy improved from approximately 76 percent to around 91 percent. This consistent upward trend in both training and validation accuracy indicates that the model was learning effectively throughout the training process. At the same time, the validation accuracy remained close to the training accuracy, which suggests that the model did not suffer from significant overfitting.

The loss values also provide insight into model performance. Training loss decreased from approximately 0.76 in the first epoch to about 0.17 in the final epoch, while validation loss decreased from approximately 0.50 to around 0.25. The reduction in both training and validation loss indicates that the model was able to minimize classification errors while maintaining stability during training.

A more detailed analysis using the confusion matrix and classification report reveals how the model performed across individual classes. The model performed exceptionally well in identifying no tumor cases, achieving a recall score of 1.00. This means that all images belonging to the no tumor category were correctly classified. The model also performed strongly in detecting pituitary tumors, with high precision and F1 scores, indicating reliable classification in that category.

However, the model showed lower performance when distinguishing between glioma and meningioma tumors. The classification report indicates that these classes had lower recall values compared to the other categories. This suggests that some images were misclassified between these two tumor types. One possible explanation for this behavior is that glioma and meningioma tumors may share similar visual characteristics in MRI images, making them more difficult for the model to differentiate. The confusion matrix shown in Figure 3 highlights the model's classification performance across all tumor categories.

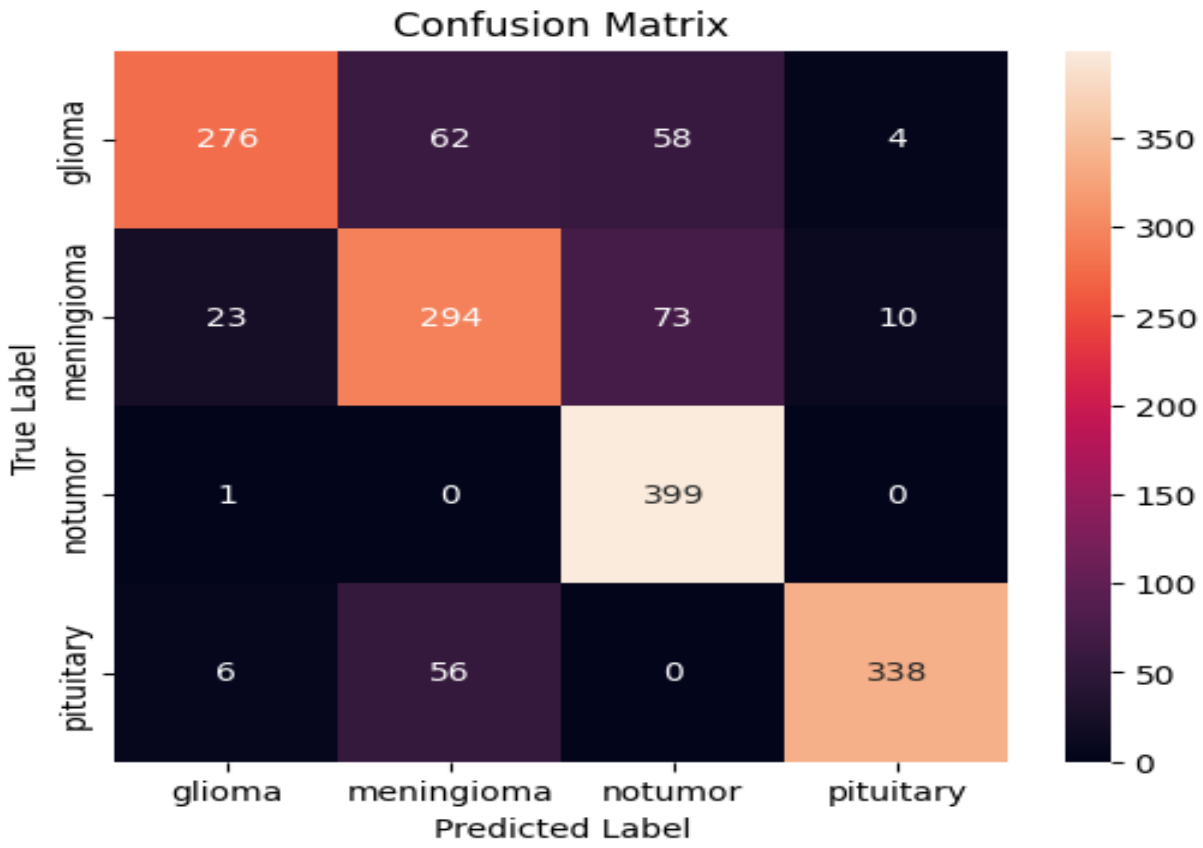

Figure 3. Confusion matrix representing the classification performance of the model across all tumor categories. The matrix highlights correct predictions along the diagonal and misclassifications in off-diagonal cells, providing insight into the model's strengths and weaknesses, particularly in distinguishing between similar tumor types such as glioma and meningioma.

Overall, the results demonstrate that the convolutional neural network is capable of performing multi-class classification of brain MRI images with a high level of accuracy. The model shows strong generalization to unseen data and performs particularly well for certain tumor types, while also highlighting areas where further improvements can be made. These results suggest that while the model performs well on the current dataset, further validation on more diverse clinical data is necessary to ensure reliability in real-world applications.

8. Limitations and Future Work

Although the model achieved strong performance, several limitations should be considered when interpreting the results. One of the primary limitations is related to the dataset used in this study. The dataset consists of labeled MRI images from a specific source, which may not fully represent the variability found in real-world medical imaging. In clinical settings, MRI images can vary significantly based on factors

such as imaging equipment, scanning protocols, patient demographics, and tumor characteristics. As a result, the model's performance may differ when applied to more diverse datasets.

Another limitation is the simplicity of the model architecture. While the use of a relatively simple convolutional neural network allowed for efficient training and easier implementation, it may have limited the model's ability to capture more complex patterns within the data. More advanced architectures, such as deeper convolutional networks or pretrained models, could potentially improve classification performance, particularly for tumor types that are visually similar.

The training process itself also introduces limitations. The model was trained for only five epochs due to time and resource constraints. Although the model showed strong improvement during these epochs, additional training could further enhance performance. Longer training, combined with careful monitoring of validation metrics, may allow the model to reach a higher level of accuracy.

In addition, the current implementation does not include advanced techniques such as data augmentation or hyperparameter tuning. Data augmentation could improve model robustness by exposing it to a wider range of variations in the input images. Similarly, optimizing hyperparameters such as learning rate, batch size, and network depth could lead to better overall performance.

Future work can address these limitations in several ways. One potential improvement is the use of transfer learning, where pretrained models such as VGG, ResNet, or EfficientNet are fine-tuned on the MRI dataset. These models have already learned rich feature representations from large image datasets and may perform better when adapted to medical imaging tasks.

Another important direction for future work is expanding the dataset to include more images from diverse sources. This would help improve the model's ability to generalize to real-world conditions. Additionally, integrating the model into a practical application, such as a decision support system for radiologists, could provide valuable insights into its real-world usability.

Overall, while the current model demonstrates strong performance, these limitations highlight opportunities for further research and improvement. Addressing these challenges can lead to more accurate, reliable, and practical AI-based systems for medical image analysis.

9. Conclusions

This project demonstrated how a convolutional neural network can be used to classify brain MRI images into glioma, meningioma, pituitary tumor, and no tumor categories. The model achieved a validation accuracy of approximately 91 percent and a test accuracy of approximately 82 percent.

The results show that even a relatively simple CNN model can learn meaningful patterns from MRI images. The model performed especially well when identifying no tumor and pituitary tumor cases. However, the confusion between glioma and meningioma images shows that further improvements are needed.

Overall, this project supports the growing role of artificial intelligence in medical image analysis. While the model is not a replacement for clinical judgment, it demonstrates the potential of deep learning as a tool for assisting healthcare professionals. This work demonstrates the practical potential of deep learning in medical imaging while emphasizing the importance of continued validation and refinement for real-world deployment.

10. References

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